Vitor Recacho

PhD Candidate · Economics

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Education

ITC Faculty of Geo-Information Science and Earth Observation

Visiting Research Fellow *2025 – present*

- Host: Dr. Adriano Barasal Morales and Dr. Luigi Lombardo

University of São Paulo

PhD in Economics *2023 – present*

- Advisor: Dr. Márcio Poletti Laurini; Co-advisor: Dr. Adriano Barasal Morales
- Thesis: Essays on Bayesian Spatio-Temporal Modeling of Streamflow and Extreme Rainfall. Defense expected November 2026

Universidade Federal do Paraná

BSc in Economics *2017 – 2021*

Professional Experience

2024–2025 **Graduate Teaching Assistant**, FEA-RP, USP

2018–2021 **Undergraduate Research Assistant**, DEST/ECONOMIA/CNPQ, UFPR

Publications

Published

Franco, J. P. M., **Recacho, V. S.**, Soares, G. C. and Laurini, M. P. (2026). A deep U-Net kriging framework for interpolating remote sensing images. *IEEE Geoscience and Remote Sensing Letters*. doi:10.1109/LGRS.2026.3708253

Ahmed, M., **Recacho, V. S.**, Titti, G., Uchida, T., Borgatti, L., Morales, A. B., Francioni, M. and Lombardo, L. (2026). On the use of rainfall time series for regional landslide prediction by means of functional regression. *Engineering Geology*, 371, 108860. doi:10.1016/j.enggeo.2026.108860

Recacho, V. S. and Laurini, M. P. (2025). Bayesian structural decomposition of streamflow time series. *Journal of Hydrology*, 649, 132478. doi:10.1016/j.jhydrol.2024.132478

In Press

Recacho, V. S. and Laurini, M. P. (in press). Spatio-temporal modeling of extreme rainfall occurrence in Brazil. *International Journal of Climatology*. doi:10.1002/joc.70671. Code: doi:10.5281/zenodo.23891518

Book Chapters

Pelaez, V. M., **Recacho, V. S.** and Monteiro, L. (2026). A inserção do Brasil no mercado internacional de agrotóxicos. In Pereira, E. M., Casazza, I. F. and Lignani, L. B. (Eds.), *História dos agrotóxicos no Brasil: usos, impactos e resistências*. Rio de Janeiro: Editora Fiocruz. doi:10.7476/9786557083062 (in Portuguese)

Under Review

Recacho, V. S., Morales, A. B. and Laurini, M. P. Detecting basin-wide hydrological change through multivariate structural streamflow modeling. *Stochastic Environmental Research and Risk Assessment* (second round).

Recacho, V. S., Morales, A. B., Lombardo, L. and Laurini, M. P. Bayesian spatio-temporal streamflow decomposition on river network metric graphs: an application to the Amazon basin. *Journal of Hydrology*.

Laurini, M. P., **Recacho, V. S.**, Morales, A. B., Lombardo, L. and Valls Pereira, P. L. Ecological heterogeneity in global warming across terrestrial biomes. *Communications Earth & Environment*.

Du, B., **Recacho, V. S.**, Wang, Y., Fang, Z., Fang, C., Morales, A. B. and Lombardo, L. Jointly estimating multiple geomorphological hazards and risks at national scales: a spatial modeling perspective. *Earth-Science Reviews*.

Du, B., Wang, Y., **Recacho, V. S.**, Tonini, M., Morales, A. B. and Lombardo, L. Over five decades of climate-related disaster losses in Switzerland: modeling losses due to hydromorphological hazards through rainfall anomalies. *International Journal of Applied Earth Observation and Geoinformation*.

Laurini, M. P., Quispe, A. D. M., **Recacho, V. S.** and Valls Pereira, P. L. Tabular foundation models for conditional income convergence. *Economics Letters*.

Zhang, Y., Du, B., **Recacho, V. S.**, Fang, Z. and Lombardo, L. Decoupling seasonal trends in landslide probability and associated intensity in the East Asian Monsoon region.

Santos, G. C. A., **Recacho, V. S.** and Laurini, M. P. From counts to extremes: a dynamic Bayesian analysis of wildfire activity in São Paulo. *Natural Hazards*.

In Preparation

Spatio-temporal natural hazard risk modeling for Switzerland.

Climate risk pricing in Brazil's subsidized rural insurance program (working title).

Bayesian modeling of wildfire durations (working title).

Awards, Fellowships, & Grants

2026	**Industrial Fellowships (IF 2026 Call): Recommended for Funding, Not Funded.**, Luxembourg National Research Fund (FNR). Project: SHIFT (Ref. 2001262), principal applicant. Recommended for funding by the Scientific Evaluation Panel
2025	**Research Computing Grant, SURF (National Compute Facilities), Netherlands.**, Project Ref: EINF-16355. Independently awarded competitive access to the **Snellius** supercomputer to support large-scale Bayesian spatio-temporal modeling and data-intensive simulations
2025–2026	**Visiting Ph.D Scholarship (Stipend)**, CAPES - Coordination for the Improvement of Higher Education Personnel. Grant number 88881.126329/2025-01
2023–2027	**Ph.D Scholarship (Stipend)**, CAPES - Coordination for the Improvement of Higher Education Personnel. Grant number 001

Presentations

Contributed Presentations

Recacho, V. S., Morales, A. B., Lombardo, L. and Laurini, M. P. (2027). Bayesian spatio-temporal streamflow decomposition on river network metric graphs: an application to the Amazon basin. Poster, VIII Latin American Meeting on Bayesian Statistics (COBAL) and XVIII Brazilian Meeting of Bayesian Statistics (EBEB), Búzios, March 2027 (accepted).

Recacho, V. S., Morales, A. B., Lombardo, L. and Laurini, M. P. (2026). Bayesian spatio-temporal streamflow decomposition on river network metric graphs: an application to the Amazon basin. Poster, Workshop on Time Series, Wavelets and High Dimensional Data (STODAD), Campinas, September 2026.

Recacho, V. S. and Laurini, M. P. (2024). Bayesian structural decomposition of streamflow time series. Oral presentation, 25th Brazilian Symposium on Probability and Statistics (SINAPE), Fortaleza.

Teaching Experience

2025/1	**Introduction to Probability and Statistics II (Undergraduate)**, Teaching Assistant	*FEA-RP/USP*
2024/2	**Econometrics III (Undergraduate)**, Teaching Assistant	*FEA-RP/USP*
2024/2	**Econometrics III (Graduate)**, Teaching Assistant	*FEA-RP/USP*
2024/1	**Applied Statistics (Graduate)**, Teaching Assistant	*FEA-RP/USP*

Outreach & Professional Development

Service and Outreach

2025–2026 **Coordination of a Postgraduate Program**, Full member
2024–2025 **Coordination of a Postgraduate Program**, Deputy member

Peer Review

International Review of Economics & Finance
Journal of Land Use Science
Regional Environmental Change

Technical Skills

Methods: Bayesian spatio-temporal modeling (latent Gaussian models, SPDE, metric graphs), point processes, extreme value theory, state-space models, machine learning
Software & Computing: R (R-INLA, inlabru), Python (PyTorch), Google Earth Engine, HPC (Snellius, SLURM)